

Aaron Lane

Biologist | Data Analyst | Former Educator

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Summary

Biologist and data analyst completing a Master's in Biological Data Science with research experience at La Jolla Institute for Immunology. Skilled in programming for data analysis and pipeline development. Published scientific author with strong communication skills. Experienced in coordinating experiments and working collaboratively across teams. Excited to apply analytical and quantitative skills to genomic analysis tools.

Experimental & Analytical Skills: Biological assay analysis, Data recording, Statistical analysis (Excel, R, Python), Experimental troubleshooting

Technical: Python, SQL, Bash scripting, Reproducible workflows, Data pipeline automation, Data cleaning and validation, Power BI

Laboratory & Research Methods: Chemistry experiment design, Data collection, Literature review, Report preparation

Communication: Scientific writing, Presentations, Team collaboration across disciplines

Additional: Adaptable and motivated to develop expertise in sequencing workflows and system integration principles

Education

M.S. Biological Data Science – Arizona State University | GPA: 4.0 | Jan 2024 - Dec 2025

M.A. Bilingual Education – University of California, Davis | GPA: 3.9 | Sept 2020 - June 2022

B.S. Biological Sciences – University of California, Santa Barbara | GPA: 3.1 | Sept 2015 - June 2019

Research & Projects

Longitudinal Flu Vaccine Response Dynamics – First Author, *Vaccine* (2025)

- Analyzed data from ~1500 individuals across 12 studies to identify immune-response phenotypes.
- Predicted long-term response trajectories with accuracy similar to assay noise; led analysis, figure design, and publication.

SERENE Mental Health App – ASU Research Project, *in progress*

- Designing a full-stack mobile application with React Native that gathers survey, image, and GPS data to study mental health–nature relationships, with cloud-based computer vision and BigQuery pipelines for automated analysis.
- Leading analysis to identify areas that need enhanced blue and green infrastructure based on mental health.

Flu Search Trends Predictor Dashboard – Independent Project

- Integrated CDC FluView and Google Trends data to predict regional influenza activity across 2021-2025 seasons.
- Built an automated Python → BigQuery → Power BI pipeline with Cloud Run scheduling to feed a live dashboard.

Heat Behind Bars – ASU Master's Capstone

- Quantified heat-related mortality risk in Arizona prisons using climate and facility infrastructure data.
- Co-authored manuscript in collaboration with ASU researchers.

Publication

Lane, A. et al. (2025). High-resolution antibody dynamics following influenza vaccination reveal predominantly weak responses as well as infrequent but durable immunity across the 2014-2022 seasons. *Vaccine*, 64, 127677.

Professional Experience

Research Assistant III – San Diego Community College District | Dec 2025 – present

- Conduct districtwide program quantitative and qualitative analyses to support institutional research on honors participation, credit transfer, and student success across colleges.
- Build and maintain Power BI dashboards with clear visualizations and narratives for administrators and policy-makers.
- Prepare, clean, and validate large enrollment and student success-datasets for state-level reporting.

Bioinformatics Specialist I – La Jolla Institute for Immunology | May 2024 – Aug 2025

- Analyzed influenza serology data to model the vaccine response; published and presented findings.
- Developed standardized analysis pipelines and collaborated across global research teams to curate large datasets.

Chemistry Teacher – Vacaville High School | Aug 2022 – Jun 2023

- Collaborated with colleagues to revamp chemistry program to include lab practicals and inquiry-based learning.
- Managed chemical storeroom, including waste storage, organization, and orders.